

Payment Ops

#	Bài toán cụ thể	Function / journey	Data signal	AI/DS/ML	Output	Được gì	Rủi ro / mất gì
1	Failed payment reason classifier	Checkout/payment failure	error code, provider response, method, bank/card, retry result	ML classification	failure reason + suggested action	Dễ làm, practical, có metric rõ	Cần synthetic/label lỗi tốt
2	Payment stuck-in-pending detector	Payment lifecycle	status sequence, time in pending, provider latency, retry count	Anomaly/rule + DS	stuck payment alert	Rất thực tế payment ops	ML có thể nhẹ nếu chỉ threshold
3	Payment status inconsistency detector	Internal vs provider status	internal_status, provider_status, webhook timestamp, last sync	Rule + anomaly	status mismatch alert	Học payment state machine	Cần mô phỏng provider state
4	Duplicate webhook detector	Webhook ingestion	event_id, payment_id, payload hash, retry count, timestamp	Rule + DS anomaly	duplicate / new event	Học idempotency/webhook cực thực tế	AI/ML nhẹ nếu đứng riêng

5	Out-of-order webhook event detector	Webhook processing	event order, event timestamp, status transition graph	Rule + sequence anomaly	invalid/out-of-order event	Rất gần vấn đề production	Cần define status machine tốt
6	Missing webhook detection	Payment waiting for async event	payment created time, expected provider event, polling result	Anomaly/rule	missing webhook alert	Practical, ops-friendly	Cần simulate timing
7	Webhook retry storm detection	Provider retry behavior	repeated webhook count, same event_id, response code, time window	DS anomaly	retry storm alert	Học observability/payment reliability	Dữ liệu synthetic
8	Idempotency key conflict detector	Payment creation/retry	idempotency_key, request hash, user_id, payment_id	Rule + anomaly	duplicate/conflicting request	Học money-movement safety	ML không rõ, nên làm module phụ
9	Double-submit payment detector	User bấm thanh toán nhiều lần	same user/order/amount/time window, idempotency key	Rule + ML-lite	possible duplicate payment	Rất thực tế checkout	Dễ trùng fraud nếu định vị sai

10	Payment retry recommendation	Sau failed/stuck payment	failure reason, provider, method, retry history, time since failure	Rule + ML ranking	retry now / retry later / stop / manual review	Có decision output rõ	Cần policy giả lập
11	Retry success prediction	Payment retry flow	previous error code, retry count, provider status, method	ML classification	retry_success_probability	ML rõ, practical	Cần labels synthetic
12	Payment method fallback recommendation	Failed checkout	failed method, user available methods, historical success	Ranking/recommendation	suggest bank transfer / wallet / card	Hợp product + ops	Cần giả lập payment methods
13	Payment method performance analytics	Payment ops dashboard	success rate, failure rate, latency by method/provider	DS analytics	method performance report	Dễ demo, useful	Ít ML nếu đứng riêng
14	Provider health scoring	PSP/provider monitoring	success rate, p95 latency, timeout rate, error spikes	DS scoring/anomaly	provider health score	Rất thực tế cho payment routing	Cần time-series event data
15	Provider outage early detection	Incident monitoring	error spike, latency spike, webhook delay, failed retries	Anomaly/time-series	outage alert	Học incident/observability	Synthetic incident simulation

16	Smart payment routing	Chọn provider/payment rail	provider health, fee, success rate, method, amount	Ranking/ML/rule	best provider route	Rất fintech/payment platform	Khó hơn, cần multi-provider simulation
17	Cost-aware provider routing	Payment optimization	provider fee, success rate, SLA, transaction type	Optimization/ranking	route recommendation	Có business angle rõ	Có thể phức tạp nếu tối ưu thật
18	Latency anomaly detection	Checkout latency monitoring	response time, provider, method, region, time of day	Anomaly detection	latency anomaly alert	Hợp DS + observability	Không quá fintech nếu thiếu payment context
19	Payment SLA breach prediction	Ops monitoring	current queue, provider latency, retry backlog, time in state	ML/time-series	SLA breach risk	Có predictive ops angle	Cần giả lập SLA/backlog
20	Queue backlog prediction for payment workers	Async processing	job queue size, processing time, error rate, retry count	Time-series/forecasting	backlog forecast	Hợp backend + ops	Xa payment nếu không có domain event
21	Checkout abandonment prediction	User checkout journey	time on page, failed attempts, method chosen, amount	ML classification	abandonment risk	Hợp product/DS	Cần frontend journey data

22	Payment UX friction detector	Checkout funnel	error at step, retry, time spent, method switch	DS funnel analysis	friction point report	Có product analytics angle	Ít ML nếu không thêm prediction
23	Payment error message improvement assistant	Failed checkout UX	error code, user context, provider message	LLM structured rewrite + eval	user-friendly error message	Hợp AI/UX, practical	Nếu chỉ rewrite text thì nhẹ
24	Support ticket auto-routing for payment issues	Customer support after payment failure	ticket text, payment status, error code	NLP classification/LLM	route to payment/refund/KYC team	Hợp ops, có NLP	Ít payment-core nếu thiếu status context
25	Payment issue root-cause assistant	Support/ops review	payment events, error codes, webhook logs, ticket text	LLM + rule evidence	root cause + next step	Hợp AI assistant có context	Cần tránh hallucination
26	Payment case summarization	Ops reviewer mở case	event timeline, retries, provider response, support notes	LLM structured summary	concise case summary	Rất useful làm add-on	Không nên standalone
27	Payment timeline reconstruction	Debug payment issue	event logs, webhook, API requests, retries	DS/event processing	ordered payment timeline	Học observability/debugging	AI/ML nhẹ nhưng rất practical

28	Payment incident clustering	Ops incident detection	similar error codes, provider, time, merchant, method	Clustering/anomaly	incident cluster	Có DS rõ, thực tế	Cần nhiều event logs
29	Payment incident severity classification	Incident triage	affected users, volume, error rate, revenue impact	ML/rule classification	severity P0/P1/P2	Học incident tooling	Cần define severity policy
30	Auto-generated incident report	Sau incident	incident timeline, metrics, root cause, actions	LLM structured report	incident postmortem draft	Hợp AI + ops	Cần evidence grounding
31	Refund status tracker	Refund lifecycle	original payment, refund request, provider refund status, settlement delay	Rule + DS	refund pending/completed/stuck	User-facing, practical	Dễ sang reconciliation nếu quá sâu
32	Stuck refund detection	Refund ops	refund age, provider status, retry count, method	Anomaly/rule	stuck refund alert	Function nhỏ, rõ	Data phải mô phỏng
33	Refund failure reason classifier	Refund failed	provider code, payment method, bank/card, amount	ML classification	refund failure reason	Có metric rõ	Cần label synthetic

34	Partial refund validation	Refund operation	original amount, refund amount, previous refunds	Rule + anomaly	invalid/valid partial refund	Học payment rules	AI/ML nhẹ
35	Refund abuse signal monitor	Refund ops/risk	refund frequency, user/merchant ratio, repeated reason	ML/anomaly	suspicious refund pattern	Practical, giao giữa ops/risk	Lấn fraud, cần định vị refund ops
36	Chargeback/dispute case classifier	Dispute intake	dispute reason, payment info, merchant, evidence	ML/NLP classification	dispute category	Rất payment-relevant	Dataset khó
37	Dispute evidence checklist generator	Merchant dispute response	dispute type, order/payment/refund info	LLM checklist + rules	evidence checklist	Hợp AI assistant	Cần source/rule rõ
38	Chargeback risk early warning	Sau payment nhưng trước dispute	merchant history, refund rate, failed delivery signal	ML classification	chargeback risk score	Có payment risk angle	Khó data, dễ sang fraud
39	Merchant payment failure analytics	Merchant dashboard	failure rate by merchant, method, provider, amount	DS analytics	merchant failure report	Hợp B2B/payment gateway	Ít ML nếu đứng riêng

40	Merchant onboarding payment readiness check	Merchant setup	callback URL, credentials, test transaction status, webhook config	Rule + DS	readiness score	Học developer experience	Không phải AI/ML core
41	Webhook endpoint reliability scoring for merchants	Merchant integration ops	response time, 2xx rate, timeout rate, retry count	DS scoring/anomaly	merchant webhook health score	Rất thực tế B2B payment	Cần simulate merchant endpoints
42	Merchant integration error classifier	API integration support	API error logs, payload validation errors, SDK version	ML/NLP classification	integration issue type	Hợp developer support	Cần logs synthetic
43	API request anomaly monitor	Payment API gateway	request rate, endpoint, error rate, auth failures	Anomaly detection	abnormal API usage alert	Học API ops/security	Dễ trùng security/fraud
44	Payment API schema violation detector	API validation	invalid payload fields, missing params, type mismatch	Rule + ML-lite analytics	common schema error report	Hợp developer experience	AI/ML nhẹ
45	SDK error pattern analyzer	Developer integration	SDK language, error stack, endpoint, version	NLP/DS analytics	SDK issue cluster	Có DX angle	Cần synthetic support logs

46	Payment sandbox scenario generator	Developer testing	scenario templates, status flows, error codes	Generative/simulation	test cases/payment scenarios	Rất hữu ích cho DX	Nếu chỉ generator thì ít ML
47	Payment test coverage analyzer	QA/payment integration	executed scenarios, edge cases, provider errors	DS/rule analytics	coverage report	Học payment QA	Không phải ML core
48	Payment status prediction	In-flight payment	current status, time elapsed, provider, method, amount	ML/time-series	likely final status	Có predictive angle hay	Cần synthetic label
49	Authorization-to-capture mismatch detector	Card-like flow	authorized amount, captured amount, capture deadline	Rule + anomaly	mismatch/expired capture alert	Học auth/capture lifecycle	Có thể quá card-specific
50	Settlement delay predictor	Payment settled later	provider, method, bank, transaction time, holidays giả lập	ML/time-series	expected settlement time	Gần payment ops, không đi sâu reconciliation	Cần settlement labels
51	Payout failure reason classifier	Merchant payout	bank response, account info validation, amount, provider	ML classification	payout failure reason	Practical B2B payment	Data synthetic

52	Payout delay anomaly detector	Merchant payout ops	payout status, bank, provider, amount, time in state	Anomaly detection	delayed payout alert	Thực tế payment/merchant ops	Dễ sang settlement/recon
53	Payment volume forecasting	Capacity planning	historical payment count, time/day, campaign/holiday	Time-series forecasting	expected volume	Học DS ops/capacity	Không đủ fintech nếu thiếu status/errors
54	Holiday/event payment spike detection	Ops monitoring	volume spike, provider latency, campaign calendar	DS/anomaly	spike alert	Hợp ops planning	Cần event calendar giả lập
55	Payment health dashboard with anomaly explanation	Ops dashboard	success rate, failure rate, stuck count, latency	DS + LLM explanation	health report + reasons	Gộp tốt nhiều module	Nếu quá rộng thì loãng
56	Payment ops knowledge-base RAG assistant	Internal ops hỏi SOP	SOP docs, error code docs, case context	RAG + eval	grounded answer + source	Hợp AI assistant	Chỉ tốt nếu gắn với payment case

57	Runbook recommendation for payment incident	Incident response	incident type, metrics, provider, historical runbooks	Retrieval/ranking + LLM	recommended runbook	Rất ops thực tế	Cần runbook dataset giả lập
58	Payment decision audit trail	Ops/compliance logging	who changed status, reason, timestamp, old/new status	SE + DS audit analytics	audit trail / suspicious change	Có compliance mindset	ML nhẹ nếu đứng riêng
59	Manual status override anomaly detection	Backoffice payment adjustment	staff_id, amount, frequency, time, merchant	Anomaly detection	suspicious override alert	Thực tế, audit/risk angle	Nhạy cảm, cần synthetic
60	Payment ops model monitoring	Theo dõi classifier/anomaly model	prediction logs, actual issue type, drift	DS monitoring	model health dashboard	Học governance/MLOps-lite	Chỉ nên làm add-on

Package 1 — Payment Operations Intelligence Prototype

Gồm:

- #1 Failed payment reason classifier
- #2 Stuck payment detection
- #3 Payment status inconsistency detector
- #10 Payment retry recommendation

#55 Payment health dashboard

Package 2 — Webhook Reliability & Idempotency Monitor

Gồm:

- #4 Duplicate webhook detector
- #5 Out-of-order webhook event detector
- #6 Missing webhook detection
- #7 Webhook retry storm detection
- #8 Idempotency key conflict detector

Package 3 — Payment Incident & Provider Health Analytics

Gồm:

- #14 Provider health scoring
- #15 Provider outage early detection
- #18 Latency anomaly detection
- #28 Payment incident clustering
- #29 Payment incident severity classification
- #57 Runbook recommendation